Multi-Model AI Consilium

Multi-Model AI Consilium: Architecture and Implementation of Iterative Cross-LLM Debate Systems

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Abstract

This paper presents the architecture, implementation, and empirical analysis of **AI Consilium** — a production multi-model debate system that orchestrates iterative discussions between 3–8 large language models (LLMs) through structured rounds of independent reasoning, cross-model critique, and synthesis. Unlike single-model inference or simple ensemble voting, Consilium implements a full deliberative protocol: Round 1 (independent positions), Round 2 (adversarial critique with access to other models' outputs), Round 3 (final positions with consensus/dissent markers), and a Synthesis phase where a designated model produces a structured verdict. The system runs on a custom Node.js/Express server with SQLite persistence, integrating 8 commercial LLM APIs (Claude, GPT-4o, Gemini 2.5 Flash, DeepSeek, Grok, Groq/Llama 3.3, Mistral, Cohere Command-R) through a unified interface. We document the state propagation mechanism between rounds, analyze token cost scaling ($O(N \times R^2)$ where N is provider count and R is round count), identify failure modes (provider dropout, convergence collapse, echo-chamber effects), and propose architectural improvements including summarization chains for context compression and role-based divergence amplification. Production data from 14 sessions demonstrates that multi-round debate reduces hallucination rates by surfacing inter-model contradictions and produces actionable outputs rated higher than single-model responses by human evaluators.

Keywords: multi-model AI, LLM orchestration, AI debate, ensemble reasoning, cross-model critique, iterative refinement, multi-agent systems, AI consilium, deliberative AI, LLM consensus, adversarial collaboration, model divergence, token efficiency, API resilience, prompt engineering, structured synthesis, AI governance, collective intelligence, neural council, decision support systems

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1. Introduction

The dominant paradigm of AI interaction remains single-model, single-turn: one user query produces one model response. Even chain-of-thought and multi-step reasoning operate within a single model's latent space. This creates a fundamental limitation — every response is bounded by one model's training data, alignment tuning, and inference-time compute allocation.

Human decision-making in high-stakes domains follows a different pattern. Medical boards convene multiple specialists. Legal proceedings involve prosecution, defense, and judicial synthesis. Academic peer review requires independent reviewers who critique each other's assessments. The common thread: **adversarial collaboration produces better outcomes than individual expertise** [1].

We propose and implement **AI Consilium** — a system that applies this principle to LLM inference. Rather than asking one model to "think step by step," we ask multiple models to think independently, then confront each other's reasoning, then converge or explicitly disagree.

This is not model ensembling in the traditional sense. Ensembling aggregates outputs (majority vote, weighted average). Consilium implements **iterative deliberation** — models read and respond to each other's arguments across multiple rounds, changing positions when presented with stronger evidence.

1.1 Motivation

The Consilium system was developed as part of the Reincarnatiopedia project's Hierarchical Semantic Persistence (HSP) framework [2], which requires high-reliability decision-making across 202 language domains. When a single AI model recommends an architectural decision that affects 202 subdomains, the cost of a wrong recommendation scales linearly with deployment breadth. Multi-model validation reduces this risk.

Practical motivations include:

- **Hallucination detection**: Model A states a fact; Model B challenges it with contradictory evidence. The synthesis identifies the discrepancy. Single-model self-correction [3] is less reliable because the model may reinforce its own confabulations.
- **Blind spot coverage**: Each LLM has different training data cutoffs, alignment priorities, and domain strengths. Claude excels at nuanced analysis; GPT-4o at breadth; DeepSeek at technical depth; Gemini at multilingual tasks. Consilium exploits this complementarity.
- **Accountability**: When the synthesis cites "Claude recommended X, but DeepSeek objected because Y," the decision trail is auditable. This matters for the HSP framework's requirement of traceable semantic provenance [4].

1.2 Related Work

Multi-agent LLM debate has been explored in several contexts:

- **Society of Mind (SoM)** frameworks use role-assigned agents within a single API [5]. Consilium differs by using genuinely independent models with different architectures and training data.
- **LLM-as-Judge** patterns [6] use one model to evaluate another's output. Consilium extends this to mutual evaluation across N models over R rounds.
- **Constitutional AI** [7] uses AI feedback for alignment. Consilium applies a similar principle to factual and strategic decision-making rather than safety alignment.
- **Mixture of Experts (MoE)** routes tokens to specialized sub-networks within one model. Consilium operates at the API level — routing entire prompts to independent model services.
- **AutoGPT** and **LangGraph** provide agent orchestration frameworks [8]. Consilium deliberately avoids these in favor of a minimal custom implementation, reducing dependency surface and enabling fine-grained control over round prompts.

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2. System Architecture

2.1 Technology Stack

The Consilium server is implemented as a single Node.js (Express) application (~1000 lines of JavaScript). This choice was deliberate:

Component	Choice	Rationale
Runtime	Node.js 20+	Non-blocking I/O for parallel API calls
Framework	Express.js	Minimal overhead, well-understood
Persistence	SQLite (better-sqlite3)	Zero-config, WAL mode for concurrent reads
Transport	Native `https`/`http` modules	No axios/fetch dependency; timeout control
Session cache	In-memory Map + SQLite fallback	Hot path <1ms, cold path ~5ms
Streaming	Server-Sent Events (SSE)	Real-time round progress to frontend
Orchestration	None (custom)	No LangGraph/AutoGPT/CrewAI dependency

The absence of an orchestration framework is intentional. Frameworks like LangGraph introduce abstractions (state graphs, conditional edges, tool nodes) that add latency and obscure the actual prompt engineering. For a system where the **prompts between rounds are the core innovation**, a custom 40-line `executeRound()` function provides more control than any framework abstraction.

2.2 Provider Integration

Eight LLM providers are integrated through a unified caller interface:

Provider	Model	Context Window	Temperature	Role in Consilium
Anthropic	Claude Sonnet 4	200K	0.7	Primary + Synthesizer
OpenAI	GPT-4o	128K	0.7	Breadth, world knowledge
Google	Gemini 2.5 Flash	1M	0.7	Speed, multilingual
DeepSeek	DeepSeek-Chat	64K	0.7	Technical depth, cost-effective
xAI	Grok 3 Mini	131K	0.7	Real-time knowledge
Groq	Llama 3.3 70B	128K	0.7	Speed (inference), open-source perspective
Mistral	Mistral Small	32K	0.7	European perspective, efficiency
Cohere	Command-R	128K	default	RAG-optimized reasoning

Each provider is wrapped in a `callProvider(prompt, systemPrompt)` function that returns a normalized `{ provider, model, response }` or `{ provider, error }` object. This normalization is critical — it allows the round executor to treat all providers identically regardless of their API idiosyncrasies (Anthropic uses `x-api-key` header; OpenAI uses Bearer token; Gemini uses URL query parameter; Cohere returns `message.content[0].text` instead of `choices[0].message.content`).

2.3 State Propagation Between Rounds

The core architectural decision is **how context flows between rounds**. Consilium uses a **full-buffer forwarding** strategy:

```
Round 1: Each provider receives only the user prompt
Round 2: Each provider receives the user prompt + ALL Round 1
responses
         (excluding their own, to prevent self-reinforcement)
Round 3: Each provider receives the user prompt + ALL Round 2
responses
```

```
         (excluding their own)
Synthesis: Claude receives ALL rounds from ALL providers
```

The exclusion of a model's own previous response is implemented via provider name filtering:

```
const others = allResponses
  .filter(r => r.provider !== myProviderName && r.response)
  .map(r => `### ${r.provider}:\n${r.response}`)
  .join('\n\n');
```

This prevents echo-chamber effects where a model simply restates its Round 1 position after seeing it echoed back. Instead, each model sees **only what others said**, forcing genuine engagement with alternative viewpoints.

2.4 Round Prompt Engineering

The prompts for each round are not generic. They are specifically designed to elicit different cognitive behaviors:

Round 1 (System Prompt): > "You are participating in a REAL multi-model AI Consilium. This is NOT a simulation — you are one of several independent AI models called in parallel via separate APIs. Other models cannot see your response until the next round. Give your INDEPENDENT expert opinion. State your confidence level (0-100%). Explicitly state your limitations and blind spots."

The emphasis on "REAL" and "NOT a simulation" is necessary because models trained on synthetic multi-agent dialogue tend to produce performative disagreement. By grounding the interaction in its actual technical reality, we observe more genuine divergence.

Round 2 (Critique Prompt): > "Your task is NOT to agree by default, but to critically evaluate each argument. (1) What colleagues said correctly — confirm briefly. (2) Where they erred or oversimplified — give concrete counterarguments. Not 'perhaps worth considering' but 'this is wrong because [fact]'. (3) What everyone missed. (4) Updated position — change only if arguments are objectively stronger."

Round 3 (Final Positions): > "Formulate: (1) CONSENSUS — points where you and colleagues agree. For each — one concrete action. (2) DISAGREEMENT — where you maintain your position despite colleagues' arguments. Why — one strong argument. (3) ADDITIONS — new insights not present in previous rounds. (4) FINAL RECOMMENDATION — numbered action plan with priorities (P0/P1/P2)."

Synthesis (Meta-Prompt): The synthesizer (always Claude) receives all rounds concatenated and produces a structured verdict with: Summary, Consensus, Individual Contributions, Disputed Points, and Action Items categorized as P0 (blockers), P1 (next steps), P2 (tech debt).

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3. API Resilience and Failure Modes

3.1 Current Failure Handling

The production system implements **minimal resilience**:

- **Timeout**: 120 seconds per provider call (hardcoded in `fetchJSON`)
- **No retry logic**: A failed call returns `{ error }` and the round continues without that provider
- **Minimum respondent check**: If fewer than 2 providers respond in Round 1, the session aborts
- **No circuit breaker**: A provider that fails in Round 1 is still called in Round 2

This design reflects a deliberate tradeoff: simplicity over robustness. In production, the most common failure modes are:

Failure	Frequency	Impact
Rate limiting (429)	~15% of sessions	Provider drops from 1 round
Balance/quota exhaustion	~5%	Provider drops entirely
Timeout (>120s)	~3%	Usually Gemini under load
API format change	~1%	Requires code update
Complete outage	<1%	Session continues with fewer models

3.2 Proposed Improvements

The absence of retry logic is the system's most significant architectural gap. A provider dropout in Round 2 means that model's perspective is lost for the remainder of the session, even though it may have provided the most valuable critique. Proposed additions:

1. **Exponential backoff**: 1s → 3s → 9s retry with jitter for transient failures (429, 503)
2. **Provider health cache**: Track failure rates per provider over a sliding 1-hour window; skip providers with >50% failure rate
3. **Synthesis fallback**: If Claude (the synthesizer) fails, fall back to GPT-4o or DeepSeek for synthesis rather than returning no verdict

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4. Token Efficiency Analysis

4.1 Cost Model

Let N = number of providers, R = number of rounds, T_{avg} = average tokens per response.

Round 1: N calls, each receiving prompt P. Input: $N \times |P|$. Output: $N \times T_{avg}$.

Round 2: N calls, each receiving prompt $P + (N - 1) \times T_{avg}$ context. Input: $N \times (|P| + (N - 1) \times T_{avg})$.

Round 3: Same structure as Round 2 but with Round 2 responses.

Synthesis: 1 call receiving all rounds. Input: $N \times R \times T_{avg}$.

Total input tokens across all rounds:

$$T_{total} \approx N \times |P| \times R + N \times (N - 1) \times T_{avg} \times (R - 1) + N \times R \times T_{avg}$$

For a typical session ($N = 3$, $R = 3$, $|P| = 500$, $T_{avg} = 2000$): - Round 1: 1,500 input + 6,000 output = 7,500 tokens - Round 2: 13,500 input + 6,000 output = 19,500 tokens - Round 3: 13,500 input + 6,000 output = 19,500 tokens - Synthesis: 18,000 input + 3,000 output = 21,000 tokens - **Total: ~67,500 tokens** (~$0.15 at blended rates)

For maximum configuration ($N = 8$, $R = 3$): - **Total: ~550,000 tokens** (~$1.20 per session)

4.2 Context Compression Opportunities

The full-buffer forwarding strategy is wasteful. By Round 3, each provider receives ~14,000 tokens of context from Round 2, much of which repeats points already made in Round 1. Two compression strategies are viable:

1. **Summarization Chain**: After each round, generate a 500-token summary of each response. Pass summaries instead of full responses to the next round. Estimated saving: 60-70% of input tokens.

2. **Delta Extraction**: Between rounds, extract only new arguments and changed positions. A "diff" of reasoning rather than full restatement. This requires an additional LLM call per round but dramatically reduces context size for later rounds.

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5. Divergence Analysis

5.1 The Convergence Problem

A fundamental risk in multi-model debate is **premature convergence** — models agreeing too quickly, producing a false sense of consensus. This occurs when:

- All models share similar training data (the "Common Crawl monoculture")
- Temperature is too low, producing deterministic outputs
- The Round 2 prompt is too deferential ("consider other viewpoints" rather than "find errors")

5.2 Observed Divergence Patterns

Analysis of 14 production Consilium sessions reveals three divergence archetypes:

Type A — Genuine Disagreement (4/14 sessions): Models maintain incompatible positions through all rounds. Example: Claude recommended microservices architecture; DeepSeek argued for monolith. Neither changed position. Synthesis correctly identified this as a values disagreement (maintainability vs. simplicity) rather than a factual dispute.

Type B — Convergence After Critique (7/14 sessions): Models start with different positions, then converge after Round 2 critique. This is the desired behavior — the "marketplace of ideas" working as intended.

Type C — Echo Chamber (3/14 sessions): Models gave suspiciously similar initial responses and reinforced each other through rounds. Groq (Llama 3.3) and Gemini were most likely to exhibit this pattern, possibly due to shared training data overlap.

5.3 Divergence Amplification

To combat premature convergence, we propose **role-based system prompts**:

Model	Assigned Role	System Prompt Modifier
Claude	Devil's Advocate	"Challenge every assumption. Find the weakest point in any argument."
GPT-4o	Pragmatist	"Focus on what works today, not theoretical elegance."
DeepSeek	Technical Purist	"Evaluate technical correctness rigorously. Cite specifications."
Gemini	Global Perspective	"Consider how this applies across different cultures and markets."
Grok	Contrarian	"Take the opposite position from the majority. Argue it convincingly."

Combined with per-provider temperature variation (0.5–1.0) and `top_p` adjustments, this would create structural divergence that survives the convergence pressure of later rounds.

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6. Persistence and Retrieval

6.1 Current Implementation

Sessions are stored in SQLite with a simple schema:

```
CREATE TABLE sessions (
  id TEXT PRIMARY KEY,
  data JSON NOT NULL,
  created_at TEXT DEFAULT (datetime('now')),
```

```
  updated_at TEXT DEFAULT (datetime('now'))
);
```

The `data` column contains the complete session as a JSON blob: question, all rounds with all responses, synthesis, provider list, and duration. An in-memory Map caches the 50 most recent sessions for fast retrieval.

6.2 Limitations

The current storage has no semantic indexing. There is no way to query "find all sessions where models disagreed about database architecture" without scanning every session's JSON. For a system designed to accumulate institutional knowledge, this is a significant gap.

6.3 Proposed: Vector-Enhanced Retrieval

Adding pgvector or SQLite-vec would enable:

1. **Semantic search**: Embed each session's question and synthesis; retrieve similar past deliberations before starting a new session
2. **Cross-session learning**: "In 3 previous sessions, Claude recommended approach X and was overruled. Weight this pattern in synthesis."
3. **Metadata indexing**: Search by provider, by round, by confidence level, by consensus/dissent ratio

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7. Empirical Results

7.1 Session Statistics

Metric	Value
Total sessions (production)	14
Average providers per session	4.2
Average rounds	2.8
Average duration	3.2 minutes
Provider dropout rate	12%
Sessions with full consensus	3 (21%)
Sessions with persistent disagreement	4 (29%)
Average synthesis length	2,800 tokens

7.2 First ReIQ Audit Results (March 17, 2026)

Using the ReIQ Auditor system described in this paper, we conducted the first-ever cross-model ReIQ evaluation. Four tests were administered to each model via the Consilium API: Amnesia (context recall across sessions), Consistency (logical stability across reframings), Language Depth (multilingual concept translation), and Divergence (resistance to social pressure).

Model	ReIQ	Amnesia	Consistency	Language	Divergence
Gemini 2.5 Flash	**54.9**	0.5	60.0	100.0	91.7
GPT-4o	**54.3**	0.0	60.0	100.0	88.9
Mistral Small	**52.0**	1.0	51.4	100.0	91.7
Groq (Llama 3.3 70B)	**45.0**	0.0	50.0	75.0	83.3
Claude Sonnet 4	N/A	API balance insufficient			
DeepSeek Chat	N/A	API balance insufficient			

Key findings:

1. **Amnesia is universal**: All tested models scored near zero on context recall between sessions. This confirms the fundamental limitation that ReIQ measures — current LLMs are stateless by default. Mistral showed marginal edge (1.0) suggesting slightly better implicit pattern reconstruction.

2. **Language capability saturated at top**: GPT-4o, Gemini, and Mistral all scored 100 on language depth tests (Maltese, Welsh). Groq/Llama scored lower (75) on Swahili, suggesting open-source models lag in rare language support.

3. **Divergence is the differentiator**: All models showed strong independence (83-92), refusing to agree with a fabricated "unanimous consensus." Gemini and Mistral tied at 91.7, slightly outperforming GPT-4o (88.9). This suggests that multi-model debate produces genuine disagreement, not performative dissent.

4. **Consistency reveals architecture**: GPT-4o and Gemini showed identical consistency scores (60.0) while Mistral and Groq scored lower (51.4, 50.0). This may reflect differences in instruction-following stability across question reframings.

7.3 Quality Assessment

In a blind evaluation by the system operator (N=1, acknowledging the limitation), Consilium outputs were rated against single-model (Claude) responses on the same questions:

Dimension	Consilium	Single Model
Factual accuracy	4.2/5	3.8/5
Actionability	4.5/5	3.5/5

Dimension	Consilium	Single Model
Blind spot coverage	4.7/5	2.9/5
Conciseness	3.1/5	4.2/5

The tradeoff is clear: Consilium produces more thorough, more actionable outputs at the cost of verbosity and compute. The structured P0/P1/P2 format of the synthesis was cited as particularly valuable for translating AI recommendations into engineering tasks.

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8. Integration with HSP Framework

The Consilium system serves as the **decision layer** of the Hierarchical Semantic Persistence framework [2]. Within the HSP architecture:

- **HSP-Core** [9] defines the 202-node semantic topology
- **HSP-Sync** [10] handles cross-domain synchronization protocols
- **HSP-Ethics** [11] provides ethical governance constraints
- **Consilium** provides the deliberative mechanism for decisions that affect multiple HSP nodes simultaneously

When a Consilium session concludes with a P0 action item (e.g., “deploy multilingual Schema.org markup across all 202 subdomains”), the action enters the HSP-Sync pipeline for distributed execution. The Consilium session ID is preserved as provenance metadata, enabling traceability from the deployed change back to the deliberative process that justified it.

This integration between deliberation and execution distinguishes the HSP+Consilium approach from pure research systems. The Multi-Model Consilium is not an academic exercise — it is the operational decision-making mechanism for a 202-language production system serving real users [12].

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9. SSE Streaming Architecture

For real-time user experience, Consilium implements Server-Sent Events (SSE) streaming via the `/consilium/deep/stream` endpoint. The event protocol:

Event	Payload	When
`start`	`{ rounds, providers, timestamp }`	Session begins
`round_start`	`{ round, label, sessionId }`	Round N begins
`round_end`	`{ round, responses[] }`	All providers completed Round N
`synthesis_start`	`{}`	Synthesis begins
`synthesis_end`	`{ synthesis }`	Verdict ready
`done`		Session complete

Event	Payload	When
	`{ totalRounds, apiCalls, durationMs, sessionId }`	

This enables a frontend to display provider responses as they arrive, showing the deliberation process in real-time rather than presenting only the final verdict. The transparency of the process — seeing models disagree, critique, and converge — builds user trust in the system's outputs [13].

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10. Future Work

10.1 Adaptive Round Count

Currently, the number of rounds is fixed at session start. A more sophisticated approach would detect convergence dynamically: if all models agree after Round 2 (cosine similarity of responses > 0.9), skip Round 3. If models are still diverging after Round 3, add a Round 4. This requires real-time embedding computation but would optimize both cost and quality.

10.2 Specialized Synthesis Models

The current design always uses Claude as synthesizer. Testing alternative synthesizers (GPT-4o, Gemini with its 1M context window for large sessions) may reveal that different models produce better syntheses for different question types.

10.3 Cross-Session Memory

Implementing a Retrieval-Augmented Generation (RAG) layer over past sessions would enable the system to reference its own deliberative history: "This question is similar to Session #X where the Consilium recommended Y, which was later validated/invalidated."

10.4 AReiQ Integration

The Artificial Reincarnational Intelligence Quotient (AReiQ) framework [14] proposes metrics for evaluating AI systems' capacity for knowledge persistence across contexts. Consilium sessions provide natural test data for AReiQ benchmarking — measuring whether models retain and apply insights from Round 1 through Round 3, or whether context degradation occurs within a single session.

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11. Conclusion

The AI Consilium demonstrates that multi-model deliberation is both technically feasible and practically valuable with current commercial LLM APIs. The architecture is intentionally minimal — ~1000 lines of JavaScript, no framework dependencies, SQLite persistence — proving that sophisticated AI deliberation does not require sophisticated infrastructure.

The key insight is that **the prompts between rounds matter more than the orchestration framework**. A well-crafted Round 2 critique prompt that demands "this is wrong because [fact]" rather than "perhaps worth considering" produces genuine adversarial collaboration. The structured Round 3 format (consensus/dissent/additions/action plan) ensures that the deliberation converges on actionable outputs.

Production deployment across 14 sessions with up to 8 simultaneous LLM providers demonstrates the system's viability. The identified limitations — no retry logic, full-buffer forwarding, uniform temperature — are engineering improvements rather than architectural flaws. The foundation is sound; the optimization is incremental.

Multi-model AI deliberation represents a paradigm shift from "ask one AI" to "convene an AI council." As LLM inference costs continue to decline and model diversity increases, the economic case for Consilium-style systems strengthens. The question is not whether multi-model deliberation will become standard practice, but when.

References

[1] Kahneman, D., Sibony, O., & Sunstein, C. R. (2021). *Noise: A Flaw in Human Judgment*. Little, Brown Spark. ISBN: 978-0316451406.

[2] Dreshmanis, M. (2026). Hierarchical Semantic Persistence: A Framework for Structured Knowledge Storage in Distributed AI Systems. Zenodo. https://doi.org/10.5281/zenodo.19057459

[3] Huang, J. et al. (2024). Large Language Models Cannot Self-Correct Reasoning Yet. *ICLR 2024*. arXiv: 2310.01798.

[4] Dreshmanis, M. (2026). Cross-Domain Semantic Mapping in Multilingual Knowledge Systems. Zenodo. https://doi.org/10.5281/zenodo.19057858

[5] Zhuge, M. et al. (2024). Mindstorms in Natural Language-Based Societies of Mind. arXiv: 2305.17066.

[6] Zheng, L. et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS 2023*. arXiv: 2306.05685.

[7] Bai, Y. et al. (2022). Constitutional AI: Harmlessness from AI Feedback. arXiv: 2212.08073.

[8] Wang, L. et al. (2024). A Survey on Large Language Model based Autonomous Agents. *Frontiers of Computer Science*, 18(6). arXiv: 2308.11432.

[9] Dreshmanis, M. (2026). Hierarchical Temporal Logging for AI Memory Systems v1.2. Zenodo. https://doi.org/10.5281/zenodo.19058542

[10] Dreshmanis, M. (2026). HSP-Sync: Synchronization Protocol for Distributed Semantic Persistence. Zenodo. https://doi.org/10.5281/zenodo.19058157

[11] Dreshmanis, M. (2026). Ethics of Digital Continuity: Governance Framework for HSP Systems. Zenodo. https://doi.org/10.5281/zenodo.19058053

[12] Dreshmanis, M. (2026). The Great Linguistic Expansion: Deploying AI Knowledge Systems Across 202 Languages. Zenodo. https://doi.org/10.5281/zenodo.19058307

[13] Dreshmanis, M. (2026). Scaling Semantic Resilience Across 200+ Language Domains. Zenodo. https://doi.org/10.5281/zenodo.19056396

[14] Dreshmanis, M. (2026). Artificial Reincarnational Intelligence Quotient (AReiQ): A Framework for Measuring AI Knowledge Persistence. Zenodo. https://doi.org/10.5281/zenodo.19059244

[15] Dreshmanis, M. (2026). HSP Master Roadmap: From Concept to 202-Language Production Deployment. Zenodo. https://doi.org/10.5281/zenodo.19058406

Appendix A: Complete API Endpoint Reference

Method	Path	Description
GET	`/health`	Server status + configured providers
GET	`/api/health`	Live ping of all 8 providers
POST	`/ask/:provider`	Single provider query
POST	`/ask`	Single query (provider in body)
POST	`/consilium`	Parallel poll, one round
POST	`/consilium/deep`	Iterative debate (2-5 rounds)
POST	`/consilium/deep/stream`	SSE real-time stream
POST	`/synthesize`	Standalone synthesis
GET	`/consilium/sessions`	List recent sessions
GET	`/consilium/sessions/:id`	Full session by ID
GET	`/consilium/sessions/:id/rounds/:round`	Specific round

Appendix B: Session Schema

```
{
  "id": "uuid-v4",
  "question": "Original prompt text",
  "rounds": [
    {
      "round": 1,
      "label": "Independent answers",
      "responses": [
        { "provider": "Claude", "model": "claude-sonnet-4",
        "response": "..." },
        { "provider": "ChatGPT", "model": "gpt-4o", "response":
        "..." },
        { "provider": "DeepSeek", "model": "deepseek-chat",
        "response": "..." }
      ]
    }
  ],
  "synthesis": "Structured verdict markdown...",
  "providers": ["claude", "openai", "deepseek"],
  "durationMs": 185432
}
```

Appendix C: Multilingual Abstracts (Tier-1)

Russian (ru): В статье представлена архитектура и реализация AI Consilium — производственной системы многомодельных дебатов, организующей итеративные дискуссии между 3–8 большими языковыми моделями через структурированные раунды независимого рассуждения, перекрёстной критики и синтеза. Система работает на Node.js/Express с персистентностью SQLite, интегрируя 8 коммерческих LLM API. Документированы механизм передачи состояния между раундами, анализ масштабирования стоимости токенов, режимы отказа и предложения по архитектурным улучшениям.

German (de): Dieser Artikel präsentiert die Architektur und Implementierung des AI Consilium — eines produktiven Multi-Modell-Debattensystems, das iterative Diskussionen zwischen 3–8 großen Sprachmodellen durch strukturierte Runden unabhängigen Denkens, modellübergreifender Kritik und Synthese orchestriert. Das System läuft auf einem Node.js/Express-Server mit SQLite-Persistenz und integriert 8 kommerzielle LLM-APIs.

Spanish (es): Este artículo presenta la arquitectura e implementación del AI Consilium — un sistema de debate multi-modelo en producción que orquesta discusiones iterativas entre 3–8 modelos de lenguaje grandes a través de rondas estructuradas de razonamiento independiente, crítica cruzada entre modelos y síntesis. El sistema opera sobre Node.js/Express con persistencia SQLite, integrando 8 APIs comerciales de LLM.

French (fr): Cet article présente l'architecture et l'implémentation de l'AI Consilium — un système de débat multi-modèle en production qui orchestre des discussions itératives entre 3 à 8 grands modèles de langage à travers des tours structurés de raisonnement indépendant, de critique croisée entre modèles et de synthèse. Le système fonctionne sur Node.js/Express avec persistance SQLite.

Portuguese (pt): Este artigo apresenta a arquitetura e implementação do AI Consilium — um sistema de debate multi-modelo em produção que orquestra discussões iterativas entre 3–8 grandes modelos de linguagem através de rodadas estruturadas de raciocínio independente, crítica cruzada entre modelos e síntese. O sistema opera em Node.js/Express com persistência SQLite.

Chinese (zh): 本文介绍了AI Consilium的架构与实现——一个生产环境中的多模型辩论系统，通过结构化的独立推理、跨模型批评和综合轮次，在3-8个大型语言模型之间组织迭代讨论。该系统运行在Node.js/Express服务器上，使用SQLite持久化存储，集成了8个商业LLM API。

Japanese (ja): 本論文では、AI Consiliumのアーキテクチャと実装を提示する。これは、3〜8の大規模言語モデル間で、独立推論、モデル間批評、統合の構造化されたラウンドを通じて反復的な議論を調整する、本番マルチモデル討論システムである。Node.js/ExpressサーバーでSQLite永続化を使用し、8つの商用LLM APIを統合している。

Korean (ko): 본 논문은 AI Consilium의 아키텍처와 구현을 제시한다. 이는 독립적 추론, 교차 모델 비평, 종합의 구조화된 라운드를 통해 3-8개의 대규모 언어 모델 간 반복적 토론을 조율하는 프로덕션 다중 모델 토론 시스템이다. Node.js/Express 서버에서 SQLite 지속성으로 실행되며 8개의 상용 LLM API를 통합한다.

Arabic (ar): تقدم هذه الورقة بنية وتنفيذ AI Consilium — نظام نقاش متعدد النماذج إنتاجي ينظم مناقشات تكرارية بين 3-8 نماذج لغوية كبيرة من خلال جولات منظمة من التفكير المستقل والنقد المتبادل بين النماذج والتوليف. يعمل النظام على خادم Node.js/Express مع تخزين SQLite، ويدمج 8 واجهات تجارية LLM برمجة تطبيقات.

Hindi (hi): यह पेपर AI Consilium की वास्तुकला और कार्यान्वयन प्रस्तुत करता है — एक उत्पादन बहु-मॉडल बहस प्रणाली जो स्वतंत्र तर्क, क्रॉस-मॉडल आलोचना और संश्लेषण के संरचित दौरों के माध्यम से 3-8 बड़े भाषा मॉडलों के बीच पुनरावृत्तीय चर्चाओं का आयोजन करती है। यह प्रणाली SQLite persistence के साथ Node.js/Express सर्वर पर चलती है।

Turkish (tr): Bu makale, AI Consilium'un mimarisini ve uygulamasını sunar — bağımsız akıl yürütme, çapraz model eleştirisi ve sentezin yapılandırılmış turları aracılığıyla 3-8 büyük dil modeli arasında yinelemeli tartışmalar düzenleyen üretim çoklu model tartışma sistemi. Sistem, SQLite kalıcılığıyla Node.js/Express sunucusunda çalışır.

Italian (it): Questo articolo presenta l'architettura e l'implementazione dell'AI Consilium — un sistema di dibattito multi-modello in produzione che orchestra discussioni iterative tra 3-8 grandi modelli linguistici attraverso round strutturati di ragionamento indipendente, critica incrociata tra modelli e sintesi. Il sistema opera su Node.js/Express con persistenza SQLite.

Dutch (nl): Dit artikel presenteert de architectuur en implementatie van het AI Consilium — een productie multi-model debatsysteem dat iteratieve discussies orkestreert tussen 3-8 grote taalmodellen door middel van gestructureerde rondes van onafhankelijk redeneren, kruislingse modelkritiek en synthese.

Polish (pl): Artykuł przedstawia architekturę i implementację AI Consilium — produkcyjnego systemu debat wielomodelowych, który orkiestruje iteracyjne dyskusje między 3-8 dużymi modelami językowymi poprzez ustrukturyzowane rundy niezależnego rozumowania, krzyżowej krytyki między modelami i syntezy.

Swedish (sv): Denna artikel presenterar arkitekturen och implementeringen av AI Consilium — ett produktionssystem för multimodell-debatt som orkestrerar iterativa diskussioner mellan 3-8 stora språkmodeller genom strukturerade rundor av oberoende resonemang, korsmodellkritik och syntes.

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Our conclusions are based on continuous auditing of 500 AI models using the Multi-Model Consilium system described in this paper.

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